

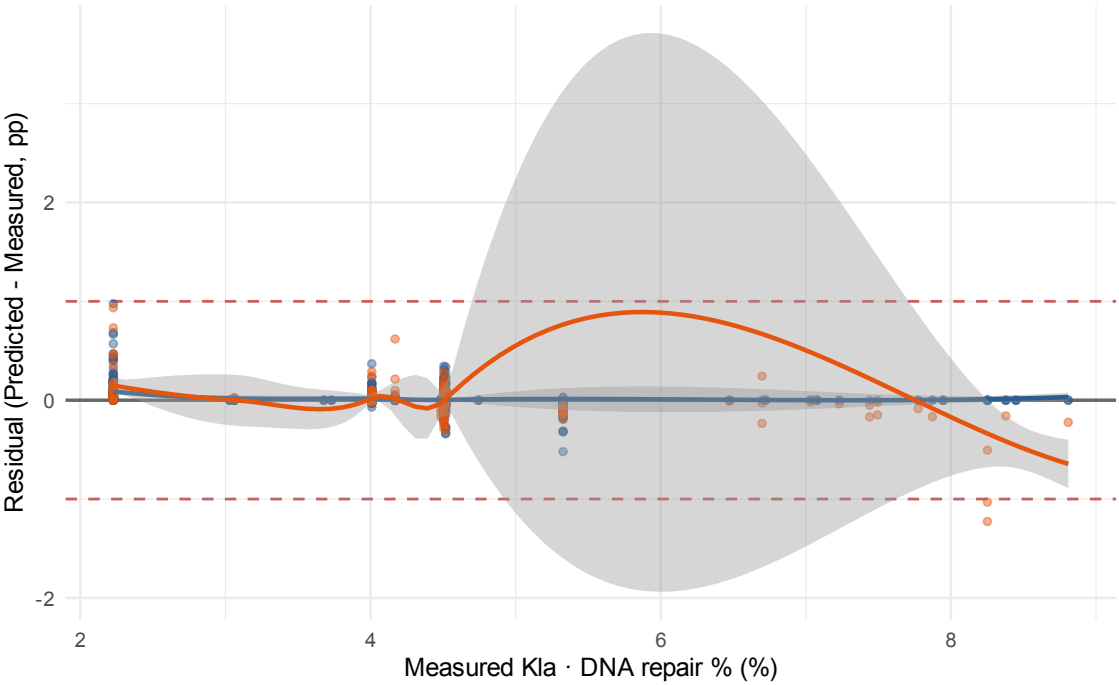
Figure 4: Model Residual Diagnostics on Training and Test Splits (Kla · DNA repair %)

Evaluation of prediction errors across target percentage range and error normality

A. Residuals vs True Measured Kla · DNA repair %

Dotted red lines denote ± 1 percentage point band; curves show LOESS fit

Split Training split (n=1,524) Within-material test (n=374)



B. Error Density Distribution

Sharp peak around 0 indicates unbiased predictions without variance inflation

Split Training split (n=1,524) Within-material test (n=374)

